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KOREA BASIC SCIENCE INSTITUTE

Overview of past/current activities in KUTC

KUTC
المركز الكوري الإماراتي المشترك للبحث والتطوير
KOREA-UAE Joint R&D Technical Center
한-UAE 공동 R&D 기술센터

December 15, 2022

Khalifa University of Science and Technology

KU Director, Daniel Choi

https://www.ku.ac.ae/kutc

دائرة التنمية الاقتصادية
DEPARTMENT OF ECONOMIC DEVELOPMENT

Ministry of Science and ICT

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Background

National Agenda

During the Korea President Moon's visit to UAE in March 2018, **both leaderships have agreed to launch a Korea-UAE Joint R&D Centre at Khalifa University** of Science, Technology and Research, through which industry professionals and academics from two countries will meet for research and discussions. Both leaderships hope experts at the centre will be able to produce meaningful outcomes in their research for our two countries' shared growth in the areas of new and renewable energy, space and bio-health.

Gulf News, *March 24, 2018*

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History of Collaboration between KU and KBSI

2013

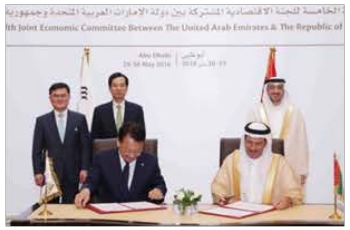
KBSI and PI proposed collaborative research to ADNOC research program

RESEARCH & DEVELOPMENT OIL SUB-COMMITTEE

ADMA OPCO

March 2015

KBSI and MI signed MOU for joint research and cooperation.



May 2016

Collaboration between KBSI and PI, MI were recognized at The 5th Economic Committee Between UAE & Korea(EC), held at Abu Dhabi.

Nov 2018

IPP candidates & jRDC will be recognized and reviewed at The 1st S&T Committee Between UAE & Korea at Abu Dhabi.

Department of Economic Development

Ministry of Science and ICT

Nov 2019

Launching Korea-UAE Joint R&D Technology Center at Khalifa University

Sept 2014

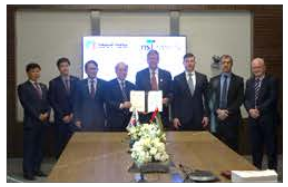
KBSI and PI signed MOU for joint research and cooperation.

Oct 2015

Dr. Hochstettler, former President of PI, his visit to KBSI opened discussion on the establishment of jRDC at PI.

March 2018

IPP & jRDC were listed on agenda at The 6th EC.
NST, on behalf of 25 GFRIs, and KU signed MOU to begin its discussion on establishment of jRDC.



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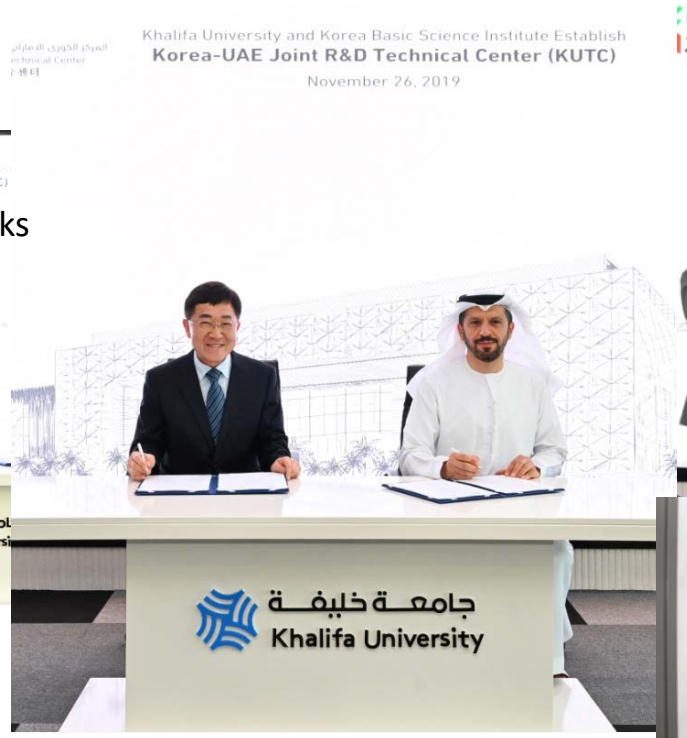
Launching KUTC on Nov. 26, 2019!

Abu Dhabi DED's congrats remarks



Signing by the leaderships (KU and KBSI)

Korea Ministry of Science's congrats remarks

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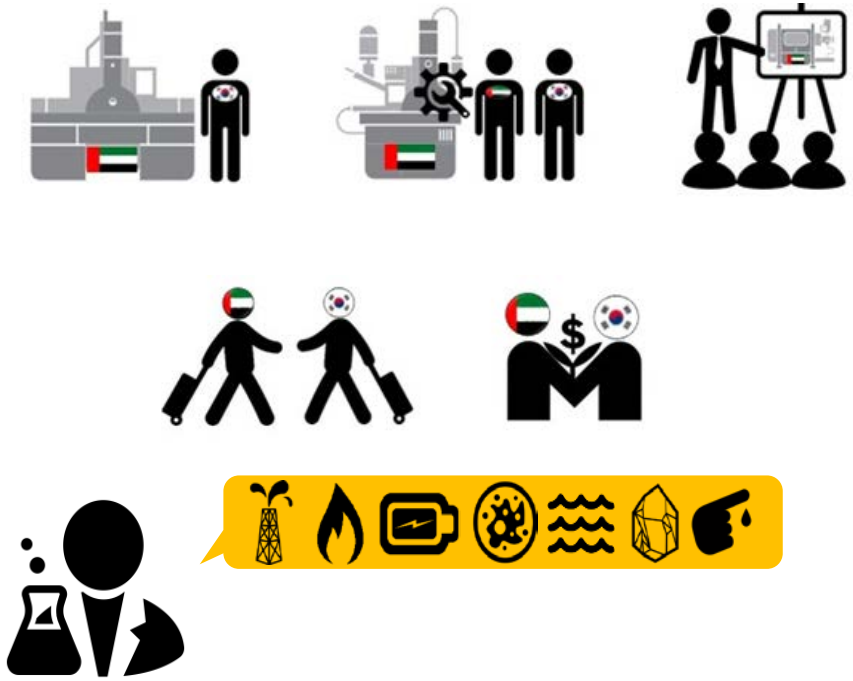
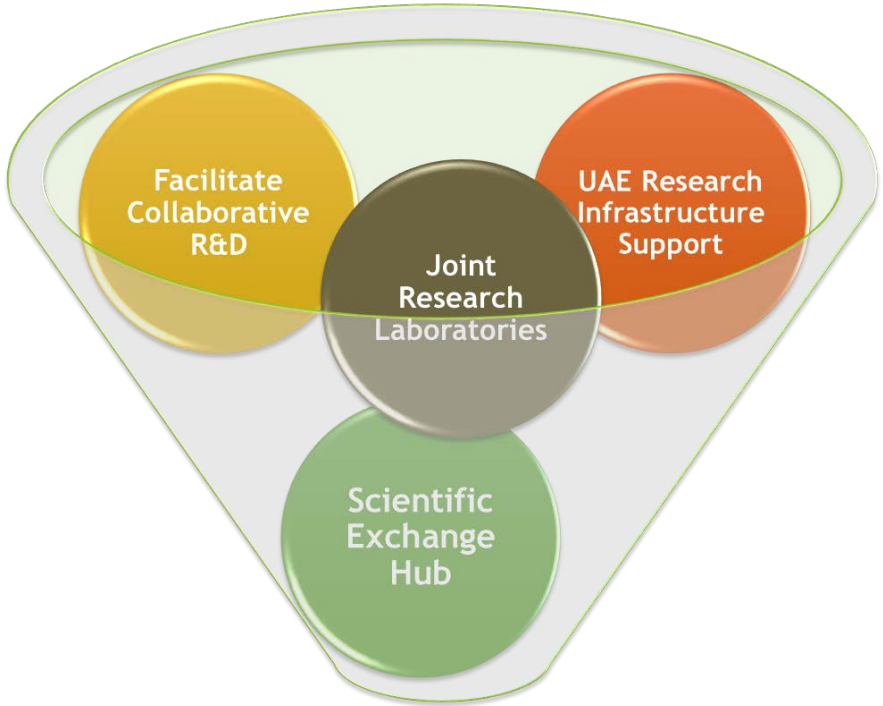
Vision and Functions

Vision: Open Research Platform in Abu Dhabi



Functions



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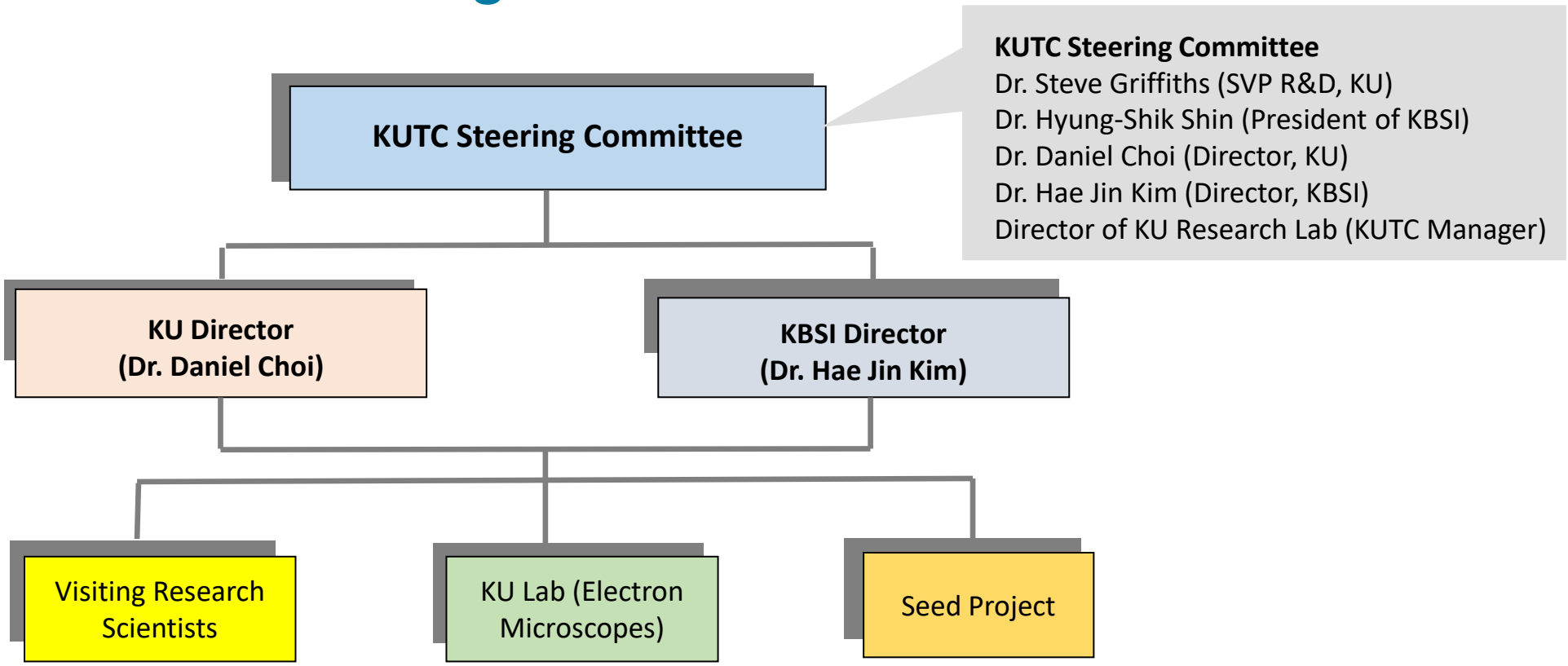
Center Organization Chart



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KUTC Joint Workshop (November 26, 2019) @ KU

Thematic Area: Advanced Materials

- Energy
- Oil & Gas
- Catalysts/ Health
- Semiconductor device/ characterization

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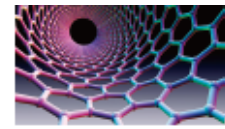

February 3, 2021

KUTC Joint Workshop 2021

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08 **Nanophotonics at KU**
Jaime Viegas

10 **Investigating the Doping of 3d Metals into Ceria by Using X-ray Dispersive Spectroscopy in a Transmission Electron Microscope Surface and interface studied with XPS**
Dalaver Hussain Anjum

12 **Inhibition of Amylin Aggregation and Toxicity**
Sung Moon Lee

14 **Mechanical behavior of 3D lattice constructed from 2D materials**
Kin Liao

16 **Mechanistic Insights of Catalytic Reactions towards a Rational Catalyst Design**
Kyriaki Polychronopoulou

18 **Bifunctional catalysts for the conversion of CO2 to hydrocarbons**
Maryam Tariq Ahmed Khaleel

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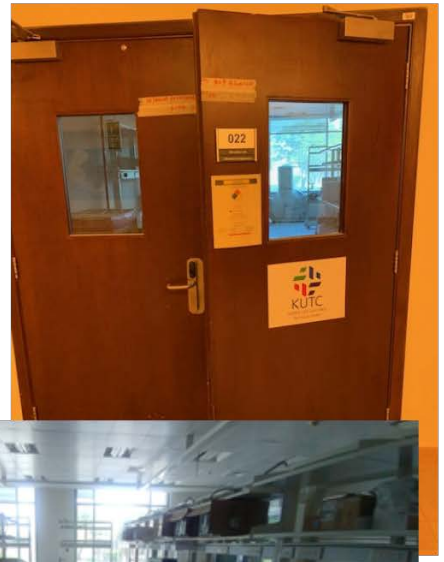
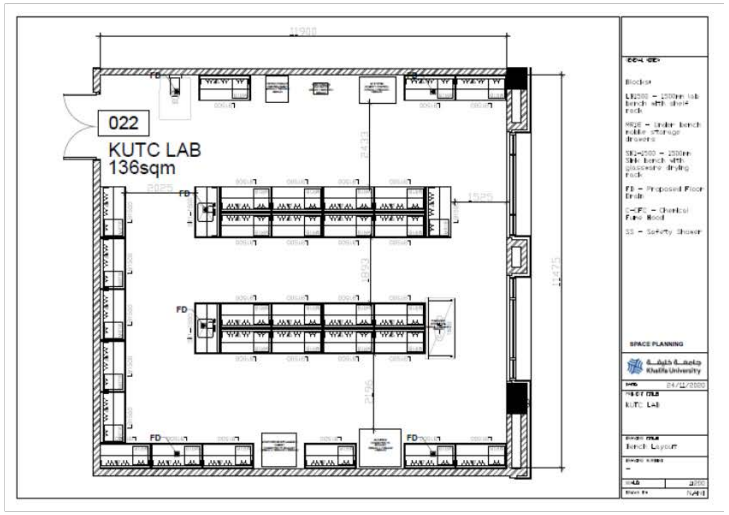
KUTC Lab

December 22, 2021

KUTC Lab Opening Ceremony, Feb 2022



8022 @ SAN – KUTC Lab



Floor Plan, 8022 – KUTC Lab

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December 22, 2021

KUTC Office @8377A SAN

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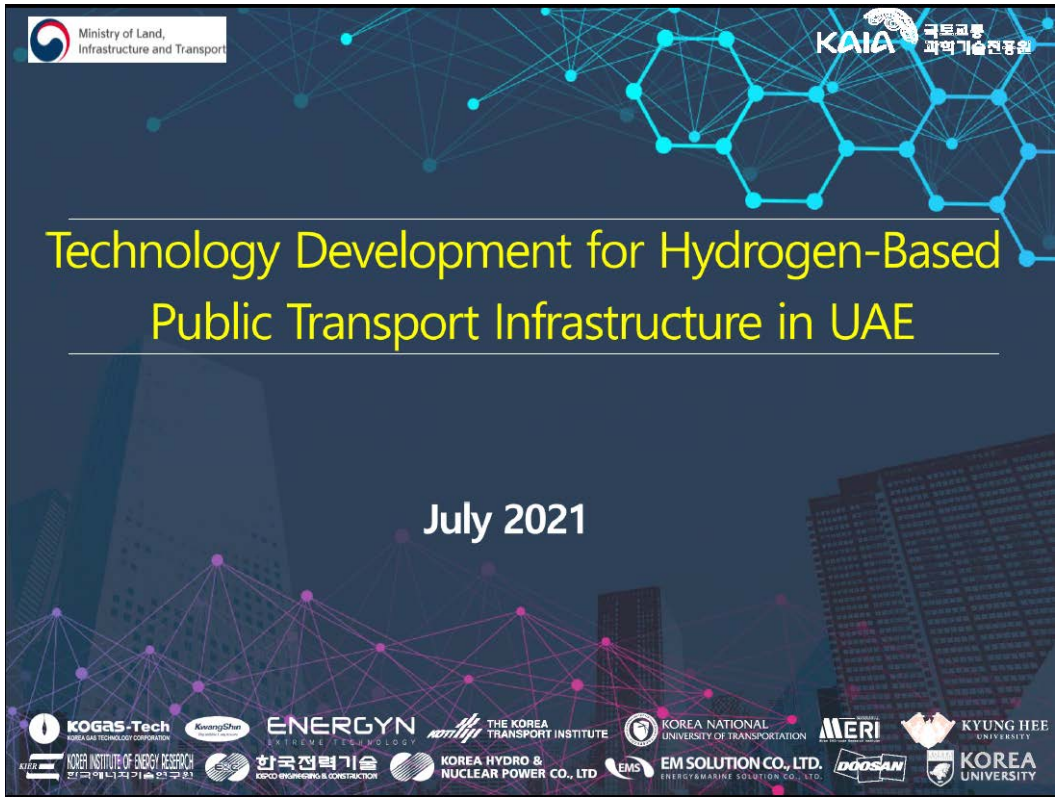
Collaboration project on Technology development for hydrogen-based transport infrastructure in UAE



January 6, 2022

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Joint Papers and IP

❑ Papers

- Joint:

1. (From KUTC Project) A, Kanagaraj, P. Chaturvedi, **H. Kim** and D. Choi, "Controllable Synthesis of $LiFePO_4$ Microrods and Its Superior Electrochemical Performance", Materials Letters, 283, 128737, 2020.
2. (From KUTC Project) Alhammadi. A., **Yun, H.J**., Choi, "Investigation of $LiFePO_4$/MWCNT cathode-based half-cell lithium-ion batteries in subzero temperature environments" Scientific reports. *(Revision in progress, 2022).*
3. (From KUTC Project) P. Chaturvedi, W. Hong, H. Kim and D. Choi, "Fabrication of flexible all solid-state Li-ion battery using thin, flexible hybrid solid polymer electrolyte in ambient condition. *(Manuscript in preparation, 2022)*
3. Nitul S Rajput, **Hionsuck Baik**, Jin-You Lu, Srinivasa Reddy Tamalampudi, Raman Sankar, Amal Al Ghaferi, Matteo Chiesa "Revealing the Quasi-Periodic Crystallographic Structure of Self-Assembled $SnTiS_3$ Misfit Compound", The Journal of Physical Chemistry C, 125, 9956, 2021.
4. I. Taha, N. Rajput, **H. Baik**, F. Ravaux, B. Alshaehhi, D. Choi, "Investigation on the interaction between GaN surface and H_2O using a nanometer-scale GaN lamella structure", Journal of Physics D: Applied Physics, 53, 465103, 2020.

- KU only:

1. (From KUTC Project) P. Chaturvedi, A. Kanagaraj, A. Alhammadi, H. Alshibli and D. Choi, "Fabrication of PVDF-HFP based microporous membranes by the tape-casting method as a separator for flexible Li-ion batteries", Bulletin of Materials Science, 44, 2, 161, 2021.
2. (From KUTC Project) P. Chaturvedi and D. Choi, "Fabrication of characterization of LTO-based CNT nanocomposites for Li-ion battery, Ionics, 2022 *(In review).*

Joint Papers and IP

- **IP**
  - Joint: None
  - KU only:

1. (From KUTC Project) P. Chaturvedi, D. Choi "Li-ion battery based on solid electrolyte flexible membrane". *US Provisional Patent application filed in September 2022, US 63/399,935*
1. (From KUTC Project) P. Chaturvedi, D. Choi "A method of fabrication of PVDF-HFR based microporous membranes as separator for Li-ion battery". (Invention Disclosure in preparation)

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Thank You